

Constant-coefficient SOS reachability: an exact classification and a Motzkin counterexample

Candidate full negative answer to the final question of arXiv:0811.4374

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Status. Candidate full counterexample and exact classification, likely valid. For every $n \geq 2$, the polynomials attainable in the source question are exactly the sums of polynomial squares. Hence every nonnegative non-SOS polynomial is a counterexample; an explicit two-variable Motzkin polynomial works in all dimensions $n \geq 2$.

1 The source question

For $n \geq 2$, let

$$\mathcal{S}(n) = \left\{ \sum_{j=1}^r g_j^2 : g_j \in \mathbb{R}[x_1, \dots, x_n], \ r < \infty \right\}$$

and let $\mathcal{N}(n)$ be the cone of polynomials nonnegative on $\mathbb{R}^n$. Borcea asks whether every $p \in \mathcal{N}(n)$ can be represented as $T(f)$, where $f \in \mathcal{S}(n)$ and T is a linear partial differential operator with constant coefficients satisfying $T(\mathcal{S}(n)) \subseteq \mathcal{N}(n)$ [1, p. 11]. The original question is reproduced in Figure 1.

so the desired conclusion follows from (3.2). □

In view of Theorem 4.4 and Corollary 4.2 one may ask whether the following analog of Hilbert's 17th problem holds, cf. [7, Problem 17].

Question. *Is it true that for any $p \in \overline{POS}(n)$ there exist $f \in SOS(n)$ and a linear partial differential operator with constant coefficients $T \in \mathfrak{L}(n)$ such that $T(SOS(n)) \subseteq \overline{POS}(n)$ and $T(f) = p$?*

Figure 1: The final question on page 11 of arXiv:0811.4374v3. The source uses $SOS(n)$ and $\overline{POS}(n)$ for the cones denoted here by $\mathcal{S}(n)$ and $\mathcal{N}(n)$.

The answer is no in every dimension under consideration. In fact, the possible outputs admit a complete description.

2 Exact reachability classification

As in the source, an operator with constant coefficients may have infinite order:

$$T = \sum_{\gamma \in \mathbb{N}_0^n} q_\gamma \partial^\gamma, \quad q_\gamma \in \mathbb{R}. \tag{1}$$

The sum is nevertheless finite on each polynomial. Define

$$\mathcal{R}_n = \{T(f) : f \in \mathcal{S}(n), T \text{ is as in (1), } T(\mathcal{S}(n)) \subseteq \mathcal{N}(n)\}.$$

Theorem 1 (automatic SOS preservation). *If a constant-coefficient operator T satisfies $T(\mathcal{S}(n)) \subseteq \mathcal{N}(n)$, then in fact*

$$T(\mathcal{S}(n)) \subseteq \mathcal{S}(n).$$

Consequently,

$$\boxed{\mathcal{R}_n = \mathcal{S}(n)}$$

for every $n \geq 1$.

Proof. Fix $d \geq 0$ and index rows and columns by $A_d = \{\alpha \in \mathbb{N}_0^n : |\alpha| \leq d\}$. Form the finite Hankel matrix

$$H_d = (h_{\alpha\beta})_{\alpha, \beta \in A_d}, \quad h_{\alpha\beta} = (\alpha + \beta)! q_{\alpha+\beta}. \quad (2)$$

For arbitrary real coefficients c_α , put $g(x) = \sum_{\alpha \in A_d} c_\alpha x^\alpha$. Since $g^2 \in \mathcal{S}(n)$, the hypothesis and evaluation at the origin give

$$0 \leq T(g^2)(0) = \sum_{\alpha, \beta \in A_d} c_\alpha c_\beta (\alpha + \beta)! q_{\alpha+\beta} = c^\top H_d c. \quad (3)$$

Thus H_d is positive semidefinite. This is precisely the constant-coefficient specialization of the Hankel condition in Theorem 4.4 of the source, but (3) also proves it directly.

Now let g be any polynomial of degree at most d , and set

$$v_\alpha(x) = \frac{\partial^\alpha g(x)}{\alpha!} \quad (\alpha \in A_d).$$

The multivariate Leibniz rule yields the polynomial identity

$$\begin{aligned} T(g^2)(x) &= \sum_{\gamma} q_\gamma \sum_{\alpha+\beta=\gamma} \frac{\gamma!}{\alpha! \beta!} \partial^\alpha g(x) \partial^\beta g(x) \\ &= v(x)^\top H_d v(x). \end{aligned} \quad (4)$$

Choose a real Gram factorization $H_d = B^\top B$. Equation (4) becomes

$$T(g^2)(x) = \sum_k \left(\sum_{\alpha \in A_d} B_{k\alpha} \frac{\partial^\alpha g(x)}{\alpha!} \right)^2, \quad (5)$$

an explicit sum of polynomial squares. Applying (5) to every summand of $f = \sum_j g_j^2$ proves $T(f) \in \mathcal{S}(n)$.

It follows that $\mathcal{R}_n \subseteq \mathcal{S}(n)$. The reverse inclusion is immediate: for $p \in \mathcal{S}(n)$, take $f = p$ and T to be the identity operator. The identity has constant coefficients and maps $\mathcal{S}(n)$ into $\mathcal{N}(n)$. $\square$

Remark 2 (GNS interpretation). Equation (2) defines a positive semidefinite polynomial moment functional $L(x^\gamma) = \gamma! q_\gamma$. Then $T(g^2)(x) = L_t(g(x+t)^2)$. Only finitely many classes of monomials occur for a fixed g ; choosing an orthonormal basis in their finite-dimensional GNS span again turns the squared Hilbert-space norm into the finite SOS decomposition (5). Thus the conclusion does not require the multivariate moment functional to possess a representing measure.

3 An explicit counterexample in every dimension

Put

$$m(x, y) = x^4y^2 + x^2y^4 + 1 - 3x^2y^2. \quad (6)$$

The arithmetic–geometric mean inequality gives

$$x^4y^2 + x^2y^4 + 1 \geq 3\sqrt[3]{x^6y^6} = 3x^2y^2,$$

so $m \geq 0$ on $\mathbb{R}^2$. We include the short standard proof that m is not SOS, making the counterexample self-contained.

Lemma 3 (Newton-polytope support restriction). *If a real polynomial P is a sum of squares $P = \sum_j Q_j^2$, then*

$$\text{Newt}(Q_j) \subseteq \tfrac{1}{2} \text{Newt}(P) \quad \text{for every } j.$$

Proof. If an exponent in some Q_j lay outside the half polytope, a separating linear weight would make twice its weight strictly larger than the maximal weight in P . But the highest-weight part of $\sum_j Q_j^2$ is the sum of squares of the highest-weight parts of the Q_j and therefore cannot cancel. This contradicts the definition of the Newton polytope of P . $\square$

Lemma 4. *The polynomial m in (6) does not belong to $\mathcal{S}(2)$.*

Proof. If $m = \sum_j g_j^2$, then $\deg g_j \leq 3$. Homogenizing each g_j to degree three would express the Motzkin form

$$M(x, y, z) = x^4y^2 + x^2y^4 + z^6 - 3x^2y^2z^2 \quad (7)$$

as a sum of squares of cubic forms.

The Newton polytope of M is the triangle with vertices $(4, 2, 0)$, $(2, 4, 0)$, and $(0, 0, 6)$; the exponent $(2, 2, 2)$ is its barycenter. The lattice points in half this triangle are exactly

$$(2, 1, 0), \quad (1, 2, 0), \quad (0, 0, 3), \quad (1, 1, 1).$$

Lemma 3 therefore forces every cubic summand to have the form

$$Q_j = a_jx^2y + b_jxy^2 + c_jz^3 + d_jxyz.$$

In $\sum_j Q_j^2$, the coefficient of $x^2y^2z^2$ is then $\sum_j d_j^2 \geq 0$: no product of two distinct displayed monomials has that exponent. In (7) the coefficient is -3 , a contradiction. $\square$

Corollary 5 (full negative answer). *For every $n \geq 2$, the answer to Borcea’s final question is negative. Explicitly, view $m(x_1, x_2)$ as a polynomial in $\mathbb{R}[x_1, \dots, x_n]$. It belongs to $\mathcal{N}(n)$, but there are no $f \in \mathcal{S}(n)$ and no constant-coefficient T with $T(\mathcal{S}(n)) \subseteq \mathcal{N}(n)$ such that $T(f) = m$.*

Proof. Theorem 1 would force $m = T(f)$ to be SOS in n variables. Setting $x_3 = \dots = x_n = 0$ in such a decomposition would make m SOS in two variables, contrary to Lemma 4. $\square$

4 Scope, verification, and novelty

The classification covers the source’s full operator class, including infinite-order constant-coefficient operators. No moment-measure theorem is used; finite positive-semidefinite Hankel matrices and their Gram factorizations suffice. The argument also isolates why allowing a nonnegative output does not enlarge the image cone: positivity on all input squares already supplies the same Gram matrix needed for an output SOS.

The Leibniz/Hankel identity, the half-Newton-polytope lattice points, and the Motzkin coefficient contradiction were independently checked by the included symbolic verifier. A second derivation through the finite-dimensional GNS construction and audits of infinite order, $n = 2$, and dimension embedding are recorded in the accompanying attempt note.

Novelty is provisional. Searches through 13 August 2026 covered the exact question, the source title and DOI, constant-coefficient SOS-preserver and Hankel/Gram variants, the local run corpus, and the four works in the source’s OpenAlex citation record. The later positivity-preserver treatments arXiv:0902.0279 and arXiv:2407.15654 were inspected directly. They discuss nonnegativity preservation and related heat-flow SOS phenomena, but the searches found no statement of Theorem 1, the reachability classification, or an answer to the displayed source question. Priority is therefore not claimed. Human review should focus first on the index matching in (4); after that, the result is finite-dimensional linear algebra.

References

- [1] J. Borcea, *Classifications of linear operators preserving elliptic, positive and non-negative polynomials*, J. Reine Angew. Math. 650 (2011), 67–82; arXiv:0811.4374v3.
- [2] T. S. Motzkin, *The arithmetic-geometric inequality*, in *Inequalities* (Proc. Sympos. Wright-Patterson Air Force Base, Ohio, 1965), Academic Press, New York, 1967, 205–224.